25900251

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBTG200, MG3DSEBBT201 -

DEVELOPMENTAL BIOLOGY AND ASSISTED REPRODUCTION TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Differentiate between vitelline membrane and zona pellucida in two points.[U] / [CO1]
   Difference-Any two points -2marks
2. Identify the life stage in women marks the end of reproductive ability. [K] / [CO2]
   Menopause -2marks
3. State how many layers are formed by delamination [U] / [CO3]
   Two/more layers- 2 marks
4. Name the prenatal procedure involves taking a small amount of amniotic fluid for genetic testing [K] / [CO4]
   Amnioscentesis-2 marks
5. Show the the main purpose of fetoscopy [K] / [CO4]
   To detect genetic disorders/visualization of fetus- 2 marks
6. Identify the role of LH in the menstrual cycle [K] / [CO1]
   Triggering ovulation/ Release of egg from the ovary//Any one function related to menstrual cycle -2marks
7. Name two organisms where parthenogenesis is a regular mode of reproduction.[K] / [CO2]
   Name any two organisms -2marks
8. Name the type of blastula lacks a cavity [U] / [CO3]
   Stereoblastula- 2 marks
9. Write one of significance of pedigree analysis? [U] / [CO4]
   Significance-Any one-2marks
10. Name the 32 celled stage in cleavage? [U] / [CO3]
    Morula- 2 marks
11. Name the cells in the seminiferous tubules provide nourishment and structural support to developing sperm?[K] / [CO1]

Sertoli cells/nurse cells/sustentacular cells -2marks.

12. Identify the first milk secreted after childbirth. [K] / [CO2]

    Colostrum -2marks [2x10 = 20]

Part B

Short Essay-Answer any 4 questions

Each question carries 5 marks

13. Describe the structural and functional significance of the notochord formed from mesoderm. [K] / [CO3]

    Notochord- introduction- 1 mark, mesoderm- introduction- 1 mark, structural significance-1.5 marks, functional significance- 1.5 marks.

14. Examine the importance of ultrasound monitoring during pregnancy. [U] / [CO3]

    Introduction- 1 mark, Importance- 4 marks (minimum 4 points)

15. Explain on placental barrier. [U] / [CO2]

    Placenta- introduction- 1 mark, placental barrier-explanation-4 marks (minimum 4 points)

16. Write in detail about spermatids [U] / [CO1]

    Spermatids- introduction- 1 mark, explanation-3 marks (minimum 3 points), diagram- 1 mark.

17. Compare fast block and slow block in polyspermy prevention [U] / [CO2]

    Polyspermy - introduction- 1 mark, fast block – 2 marks, slow block- 2 marks

18. Summarise the applications of fate maps. [U] / [CO3]

    Fate map- introduction- 1 mark, Applications- 3 marks, diagram- 1 mark

[5x4 = 20]

Part C

Essay type questions-Answer any 2 questions

Each question carries 15 marks

19. Discuss the advantages and limitations of IVF ? [U] / [CO4]

    IVF- introduction- 3 marks, steps-4 marks, advantages- 4 marks, limitations- 4 marks.

20. Explain the process of gastrulation and its significance in embryonic development?[U] / [CO3]

    Gastrulation – definition- 3 marks, embryonic development-introduction- 3 marks, process of gastrulation - explain- 4 marks, diagram – 2 marks, Significance- 3 marks.

21. Discuss the role of estrogen and progesterone in lactation and mammary gland development [K] /[CO1]

    Estrogen and progesterone – introduction-5 marks, role in lactation and mammary gland development-10 marks, (minimum 10 points)

22. Explain the significance of fertilization and the major events involved. [U] / [CO2]

    Fertilization - introduction- 3 marks, types- 2 marks, stages- major events-5 marks, diagram- 2 marks, significance- 3 marks.

[15x2 = 30]